Supporting information

for the article

Copper(I) iodide - hexamethylenetetramine complex as stable scintillator for free-standing flexible and high-resolution X-ray imaging screens

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Experimental

Materials: copper(I) iodide (CuI, >99.98%, Lankhit, Russia), urotropine ($C_6H_{12}N_4$ = HMTA, chemically pure, Ruskhim, Russia), chlorobenzene (C_6H_5Cl = CB, anhydrous, 99.8%, Sigma-Aldrich); hypophosphorous acid (H_3PO_2, ~87% aqueous solution) was synthesized from $Ca(H_2PO_2)_2$ by the addition of sulfuric acid.

Reference scintillators: Cerium-doped yttrium aluminum garnet single crystal ($Y_3Al_5O_{12}$:Ce = YAG, CRYTUR, Czech Republic) with certified light yield (LY) of 30000 photons/MeV, europium-doped calcium fluoride single crystal (CaF_2:Eu = CaF_2, Kristall JSC, Russia) LY about 20000 photons/MeV, silver-doped zinc sulfide (ZnS:Ag, Eljen Technology).

Synthesis of nanoparticles. $Cu_6I_6(HMTA)_2$ nanoparticles were synthesized by combining copper iodide and urotropine solutions. The first solution was prepared by dissolving 2.1 mmol of finely ground CuI in an aqueous solution of KI (C(KI) = 4 M, 10.5 ml H_2O), and then filtered. A solution of hexamethylenetetramine (4.2 mmol in 3.15 ml H_2O) was then added dropwise under vigorous stirring. The resulting colloidal solution of $Cu_6I_6(HMTA)_2$ particles was kept under stirring and heating at a constant temperature **T** for a time **t**. The colloidal solution underwent three washing cycles (sequential centrifugation, decantation, and redispersion in H_2O via ultrasonic treatment). The final product was dried in an oven at 60 °C overnight.

Composite free-standing films. To produce a polymer scintillation composite, finely ground scintillator powder was mixed with a concentrated solution of ethylene-vinyl acetate (EVA) in a suitable solvent (chlorobenzene or chloroform). The mixture was homogenized mechanically until the complete evaporation of the solvent. The resulting composite was placed into a preheated mold (75 °C) and mechanically molded until the desired shape was achieved. After cooling, the resulted screen was removed, preserving its shape and thickness.

Radioluminescence measurements. The scintillation properties of the obtained materials were studied under X-ray radiation from a tube with a copper anode (Siemens, type KFL, E = 8.048 keV and 8.029 keV – radiation of the Cu $K\alpha_1$ and Cu $K\alpha_2$ lines) at standard parameters – voltage of 40 kV, current of 40 mA. The spectra were recorded both with additional filtration by a Ni filter (leaving only the characteristic lines of Cu Kα1 and Cu Kα2 with E = 8.048 and 8.029 keV, respectively) and without it. The samples were placed at a 90° angle to the X-rays, the spectra were recorded in transmission geometry. The signal was collected using an optical fiber brought close to the sample. To collect the radioluminescence signal more effectively and to eliminate the influence of extraneous illumination, the measurements were carried out in a dark box, and the samples were covered with a layer of thin aluminum foil on an adhesive base ("foil tape", total

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thickness <50 μm). The spectra were recorded using a QE Pro 65000 spectrometer (Ocean Optics). The RL spectra of all samples were corrected for the spectral sensitivity of the detector and the efficiency of the diffraction grating of the spectrometer (Figure S1).

The light yields (LY) of the materials were estimated by measuring the X-ray luminescence spectra of the samples and a standard scintillator sample in the same geometry and conditions, followed by integrating the intensity of the spectra and comparing it with the integrated intensity of the standards. Full calculation procedure is described in Supporting Information.

LY Measurements and Calculations

The scintillation properties of the obtained materials were studied under X-ray radiation from a tube with a copper anode (Siemens, type KFL, E = 8.048 keV and 8.029 keV – radiation of the Cu Kα1 and Cu Kα2 lines) at standard parameters – voltage of 40 kV, current of 40 mA. The spectra were recorded both with additional filtration by a Ni filter (leaving only the characteristic lines of Cu Kα1 and Cu Kα2 with E = 8.048 and 8.029 keV, respectively) and without it. The samples were placed at an angle of 90° to the beam, the spectra were recorded in transmission geometry. The signal was collected using an optical fiber brought close to the sample. To collect the radioluminescence signal more completely and to eliminate the influence of extraneous illumination, the measurements were carried out in a dark box, and the samples were covered with a layer of thin aluminum foil on an adhesive base ("foil tape", total thickness < 30 μm). The spectra were recorded using a QE Pro 65000 spectrometer (Ocean optics).

To calculate the "apparent" light yield (LY_a) of the samples the measured radioluminescence spectra were integrated over wavelength with correction by spectral sensitivity of the detector and the efficiency of the diffraction grating of the spectrometer – function $D(\lambda)$ (Figure S1) using following equation:

$$LY_a(sample) = LY_{reference}\frac{\int I_{sample}(\lambda)\cdot D(\lambda)d\lambda}{\int I_{reference}(\lambda)\cdot D(\lambda)d\lambda}$$

Further correction (normalization) of the apparent" light yield to X-ray absorption efficiency was done by the following:

$$LY_{XAE-corr} = \frac{LY_a}{\int \Phi(E)\cdot e^{-d\cdot\mu(E)}dE}$$

where $\Phi(E)$ is energy spectrum of the X-ray tube at 40 kV, $\mu(E)$ is linear attenuation efficiency, and d is the scintillator equivalent thickness.

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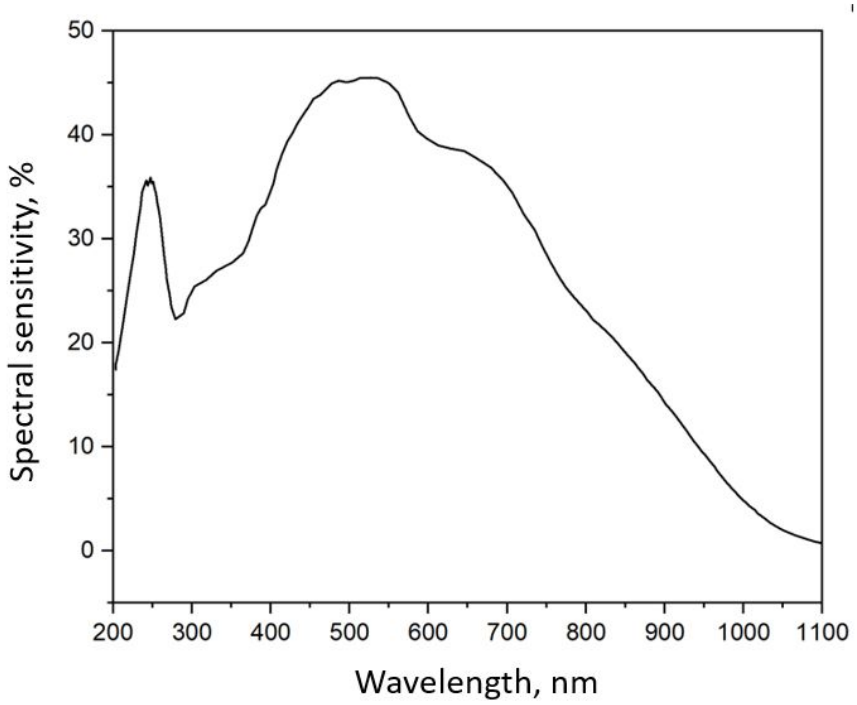


Figure S1. The sensitivity curve of the spectrometer calculated as the product of the spectral sensitivity of the detector and the wavelength-dependent efficiency of the spectrometer diffraction grating.

Photoluminescence measurements. In all the experiments to study optical properties, the spectra of the samples were recorded using powders placed between two quartz glasses to exclude glass absorption in the UV region.

Room temperature steady-state photoluminescence emission (PL) and excitation spectra (PLE) for all samples were measured on a FluoroMax Plus spectrofluorometer using a xenon lamp with a monochromator system as the excitation source. The maximum in the excitation spectra was chosen as the excitation wavelength λ_{ex} (in the range of 300–320 nm). Excitation spectra were recorded at wavelengths corresponding to the positions of the emission maxima in the PL spectra. Determination of photoluminescence quantum yields (PLQYs) for powders was carried out via the absolute method using a PTI KSPHERE-Petite integrating sphere at an excitation wavelength of 300 nm.

The modified de Mello et al. [1] method requires the measurement of the L_a – the integrated intensity of light exiting the sphere when the empty cuvette is illuminated at the excitation wavelength (Rayleigh scattering band); L_c – the same integrated intensity at the excitation wavelength when the sample is introduced into the sphere; E_a – the integrated intensity of the entire emission spectrum of the empty cuvette; and E_c – the integrated intensity of the entire emission spectrum of the cuvette with the sample. The absolute quantum yield is then given by:

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$$PLQY = \frac{E_c - E_a}{L_a - L_c} \times 100\%$$

Diffuse reflectance spectra (DRS) were recorded on a PerkinElmer Lambda 950 spectrophotometer (USA) at 298 K in the wavelength range from 370 to 800 nm.

Powder X-ray diffraction patterns were obtained using CuKα radiation on a Bruker D8 Advance diffractometer (Germany) in Bragg-Brentano geometry in the angle range of 5–45° with a step of 0.02°.

Thermogravimetric analysis was conducted using Mettler Toledo TGA 2 system (Switzerland) in a temperature range 298–900 K. A small mass of powder was placed in an alumina crucible with a pierced lid and measured in a flow of Ar (30 mL/min) at a scanning speed of 5 K/min.

Imaging and determination of spatial resolution. The images were obtained by photographing the "shadow" of a small object placed between the X-ray source and the scintillation screen. Unfiltered radiation from an X-ray tube with a copper anode at voltage of 40 kV and current of 40 mA was used as a source. For this purpose, the camera was positioned on the line of the collimated X-ray propagation, the scintillation screen was placed between the objective (Canon EF 100mm f/2.8L Macro IS USM, exposure time 1 s) and the source and the objective was focused on it, whereas the contrast object was placed in the beam in front of the screen. In this case, the object and the screen were positioned perpendicular to the X-ray beam, and the entire optical system was isolated from external light (Figure S2). Miniature flexible printed circuits were used as test contrast objects, as well as small biological objects for contrast assessment. The use of contrast objects with known geometric parameters also allows one to assess the spatial optical resolution of the manufactured scintillation screens. The maximum spatial resolution was determined using the modulation transfer function (MTF) graph, which characterizes the ability of the visualization system (scintillation screen) to reproduce an undistorted image of the object under study.

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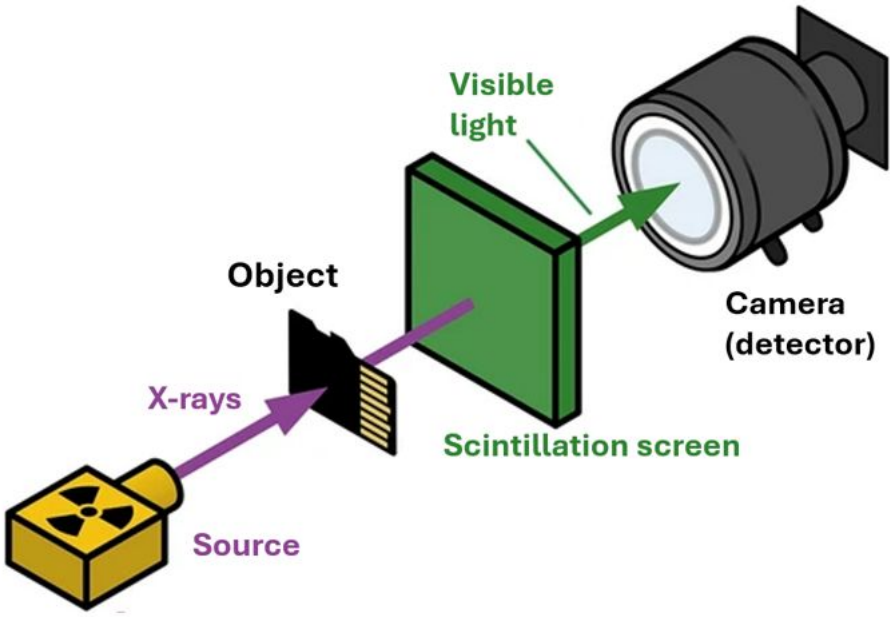


Figure S2. Scheme of the X-ray home-made imaging system.

MTF calculation procedure

As a convenient and accessible method for constructing the MTF graph, we utilized the edge method, which is based on the international standard ISO 12233 (Photography – Electronic Still Picture Imaging – Resolution and Spatial Frequency Responses, ISO 12233:2024) and is widely applied in both laboratory and industrial measurements of the resolution of various optical systems. This method requires obtaining a high-contrast image of the edge of an optically dense object with a sharp and uniform boundary – in our case, the edge of a metal blade with sufficient thickness to block X-ray transmission.

Using the acquired edge image, an edge spread function (ESF) is initially plotted with the help of image processing software (ImageJ software was used in this work). The ESF essentially represents the brightness (grey value) profile across the edge region. However, this method is generally insufficiently sensitive to minor variations in resolution. Therefore, to obtain the modulation transfer function (MTF), the ESF is further processed. Specifically, the ESF is graphically differentiated to obtain the line spread function (LSF), followed by a one-dimensional discrete Fourier transform to derive the required frequency-contrast characteristic. Origin2021 software was used for the differentiation and discrete Fourier transform of the ESF curves. To minimize random errors, LSF profiles were constructed for different overlapping and/or adjacent regions within each image (Figure S3).

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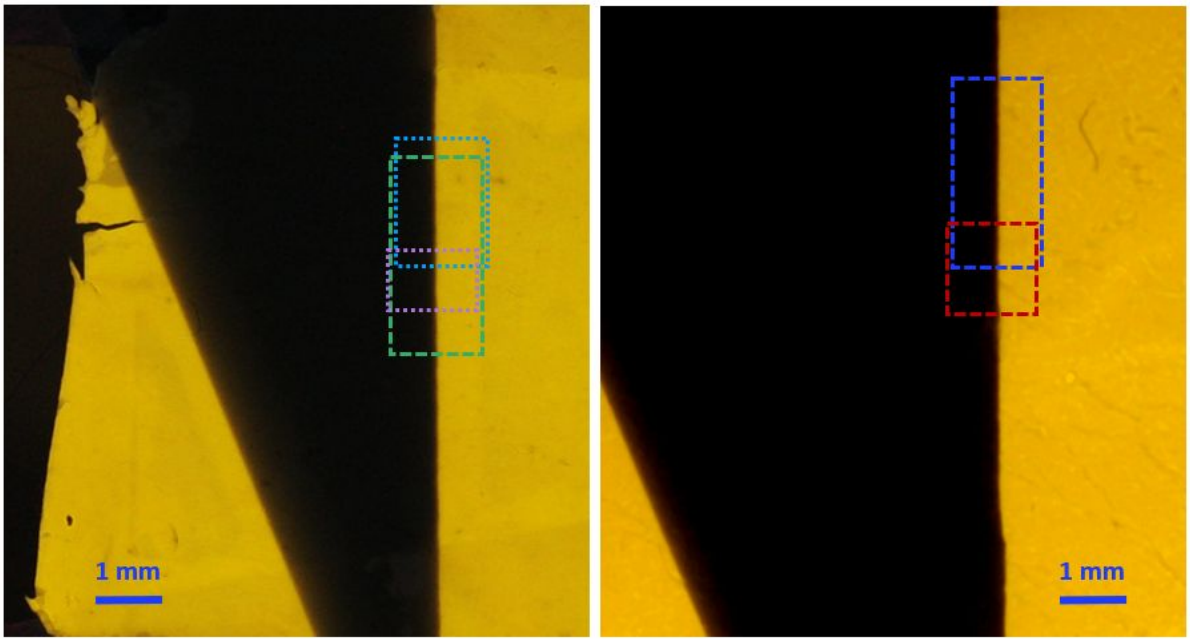


Figure S3. Photographs of the shadow under the X-ray blocking metal blade on CCS (left) and UFS (right) screens; the regions where the edge spread function was acquired are indicated by colored dashed lines.

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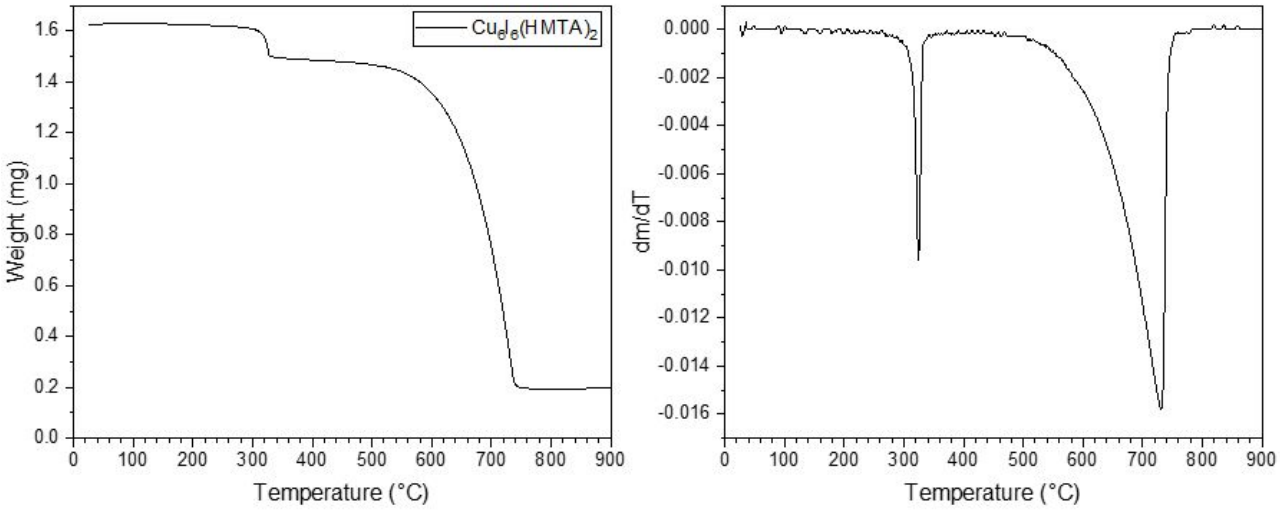
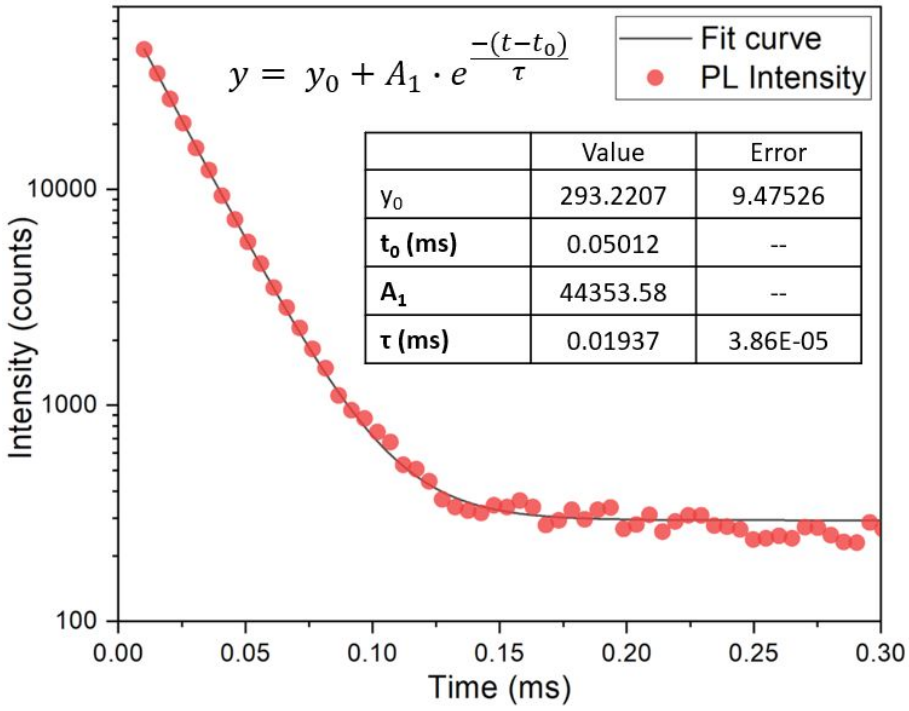


Figure S4. Thermogravimetric Analysis (TGA, on the left) and Derivative Thermogravimetry (DTG, on the right) of $Cu_6I_6(HMTA)_2$ powder.

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	Value	Error
y_0	293.2207	9.47526
t_0 (ms)	0.05012	--
A_1	44353.58	--
τ (ms)	0.01937	3.86E-05

Figure S5. Photoluminescence (PL) decay curve of $Cu_6I_6(HMTA)_2$, showing the measured PL intensity (red dots) as a function of time. The black line represents the fitted curve with a decay time of ~19.4 μs.

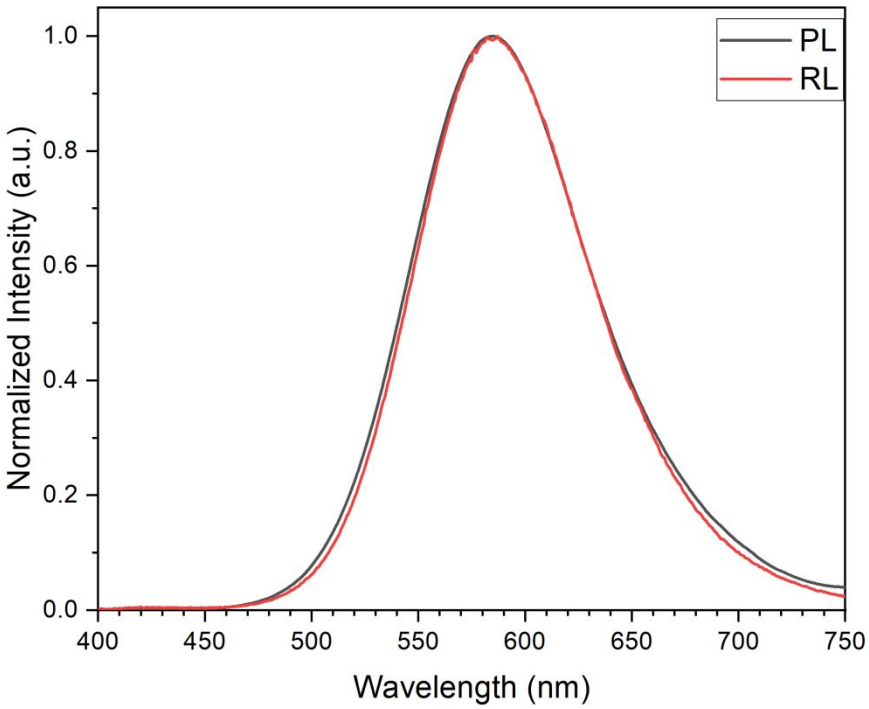


Figure S6. Normalized photoluminescence (PL) and radioluminescence (RL) spectra of $Cu_6I_6(HMTA)_2$. The black and red curves represent the PL and RL emission profiles, respectively. Both spectra exhibit a similar peak position and shape, indicating that the emission characteristics remain consistent under optical and X-ray excitation.

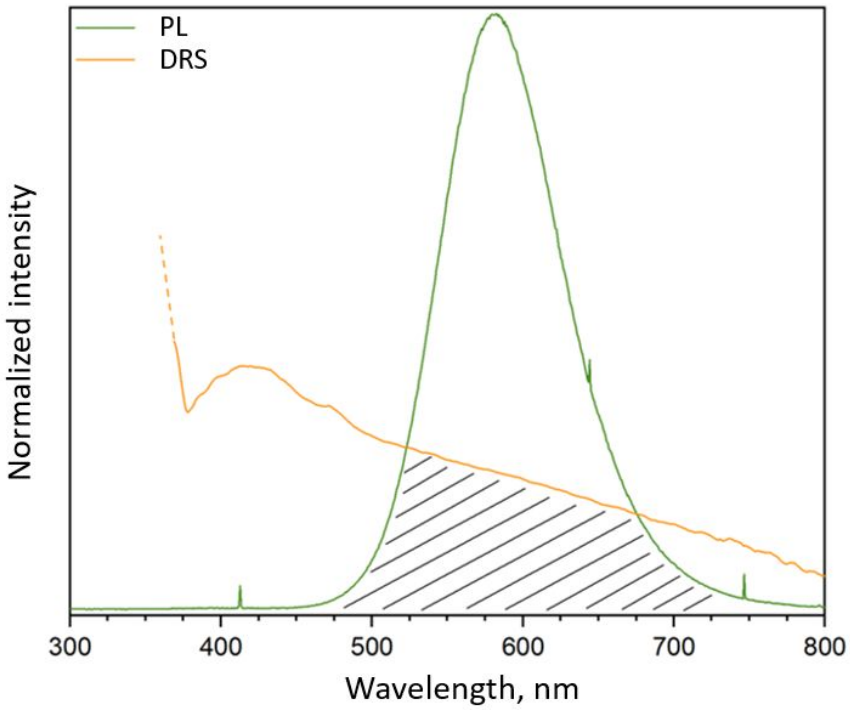


Figure S7. The overlapping of emission spectrum of $Cu_6I_6(HMTA)_2$ and its diffuse reflectance spectrum.

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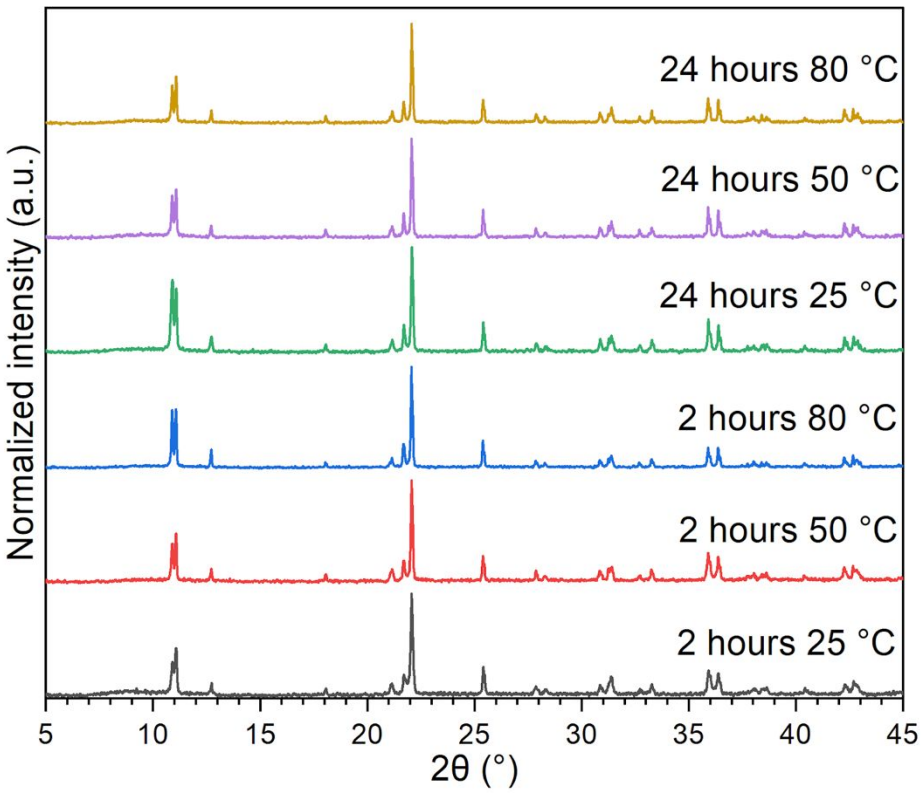


Figure S8. X-ray diffraction (XRD) patterns of $Cu_6I_6(HMTA)_2$ powders synthesized at different temperatures (25 °C, 50 °C, and 80 °C) and reaction times (2 hours and 24 hours).

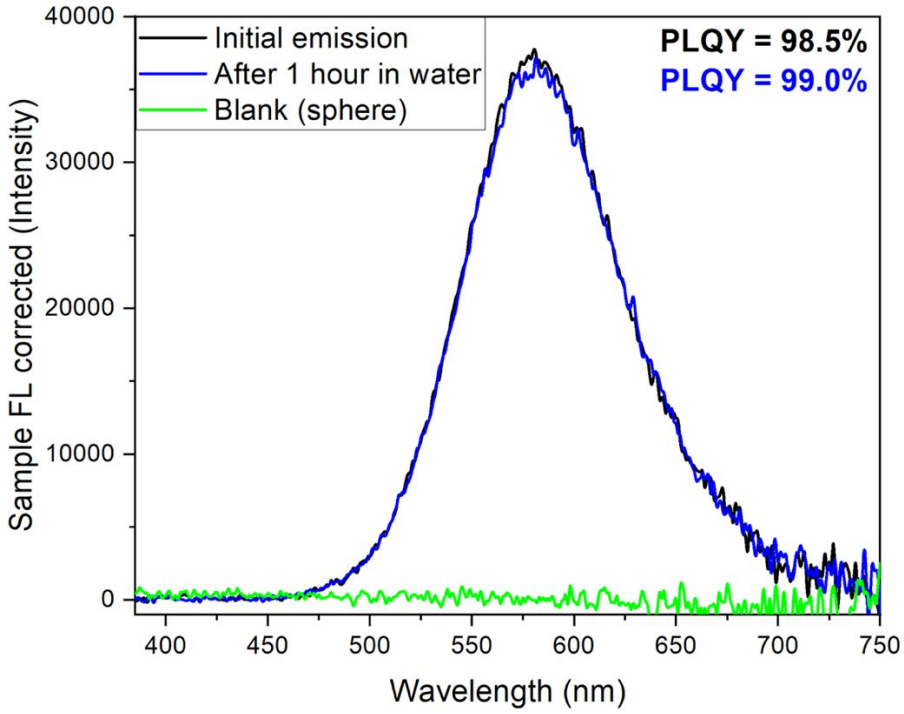


Figure S9. Corrected fluorescence (FL) spectra of the $Cu_6I_6(HMTA)_2$ composite scintillator screen, showing the initial emission (black) and the emission after one hour of water immersion (blue). The blank measurement (green) represents the background signal from the integrating sphere. The minimal change in peak position and intensity after water exposure indicates excellent moisture stability.

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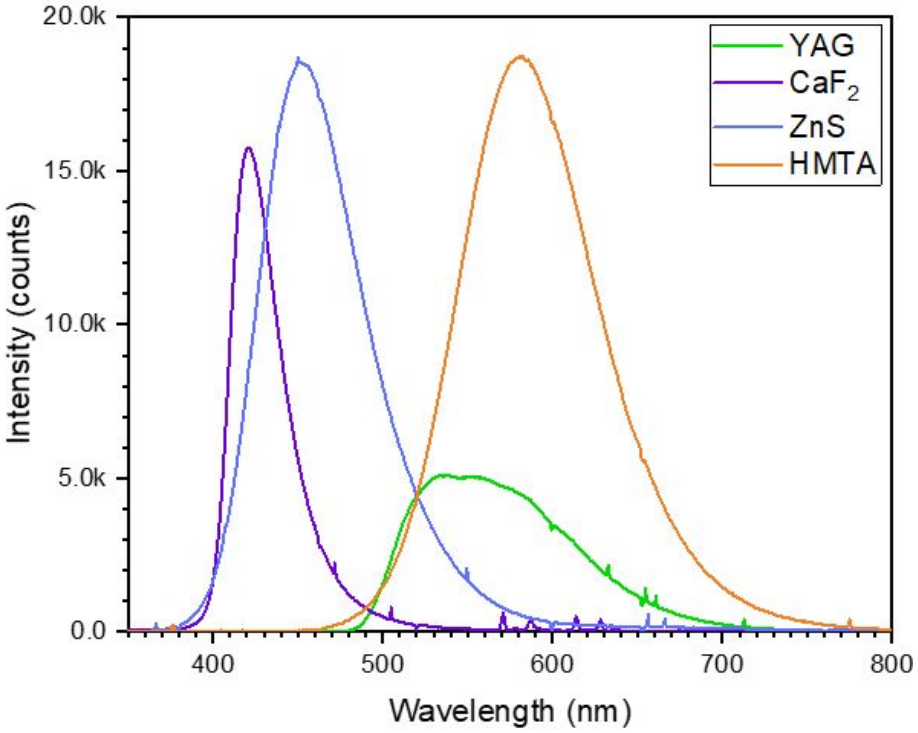


Figure S10. RL spectra of $Cu_6I_6(HMTA)_2$ composite scintillator screen (HMTA) and reference scintillators – single crystals of $Y_3Al_5O_{12}$:Ce (YAG) and CaF_2:Eu = CaF_2, and composite screens based on commercial powder of ZnS:Ag (ZnS).

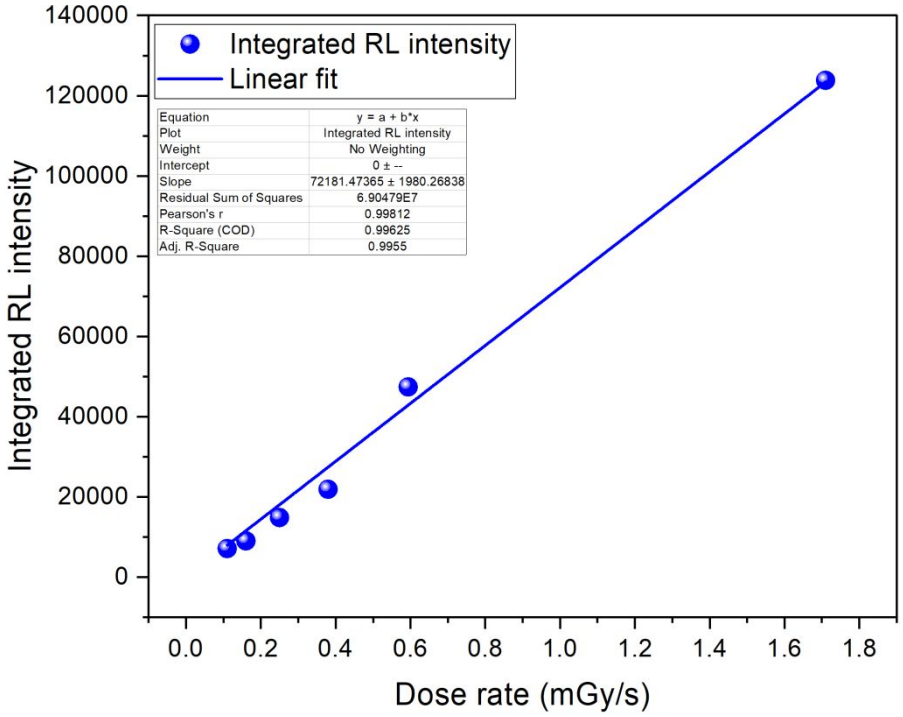


Figure S11. RL intensity dependence on the X-ray dose rate for the 100 um thick $Cu_6I_6(HMTA)_2$ composite scintillator screen.

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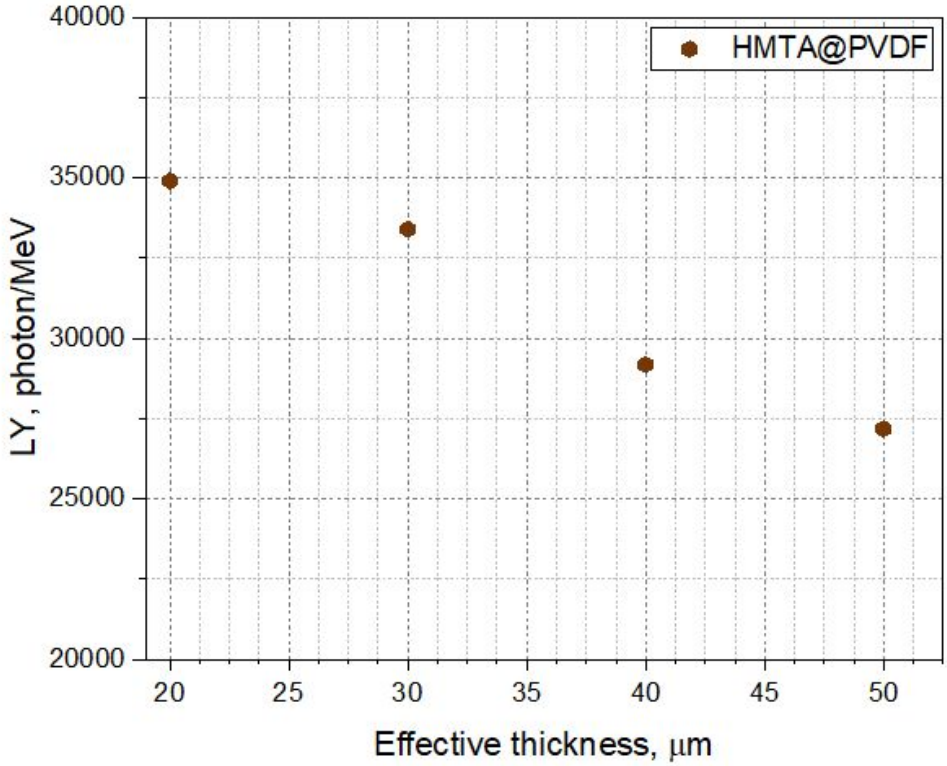


Figure S12. Dependence of the light yields of $Cu_6I_6(HMTA)_2$ ultrathin film scintillator screens (UFS) on the effective thickness for X-ray radiation (the effective thickness in the case of UFS is equal to the total thickness).

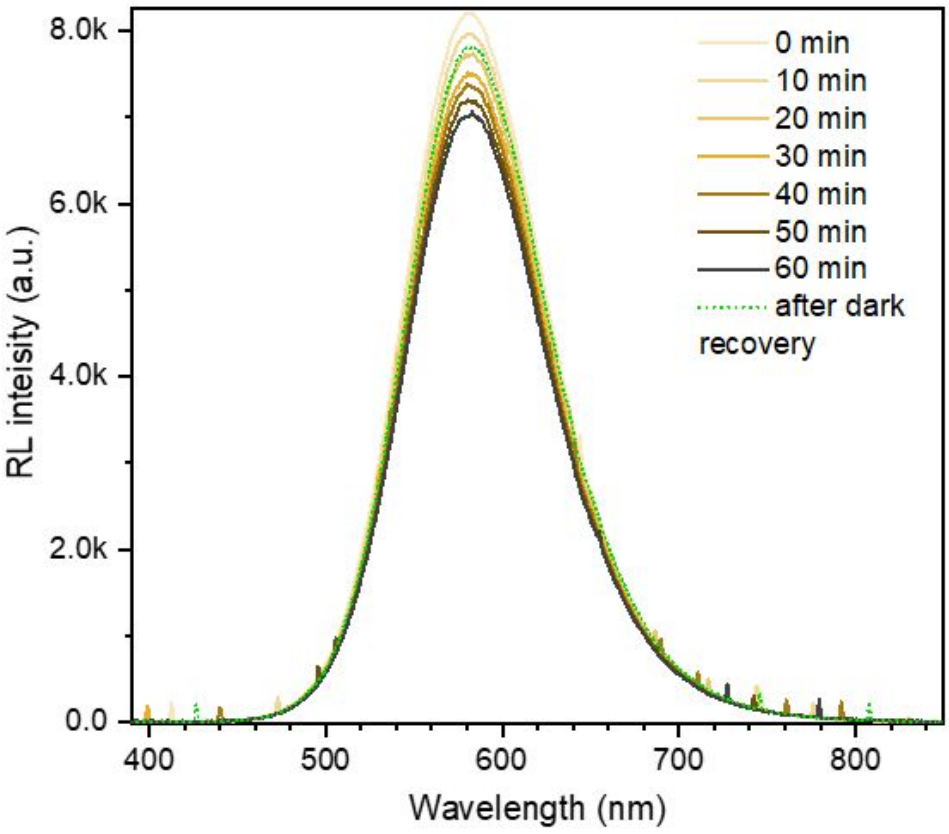


Figure S13. RL spectra of $Cu_6I_6(HMTA)_2$ CCS during X-ray exposure and after recovery in the dark.

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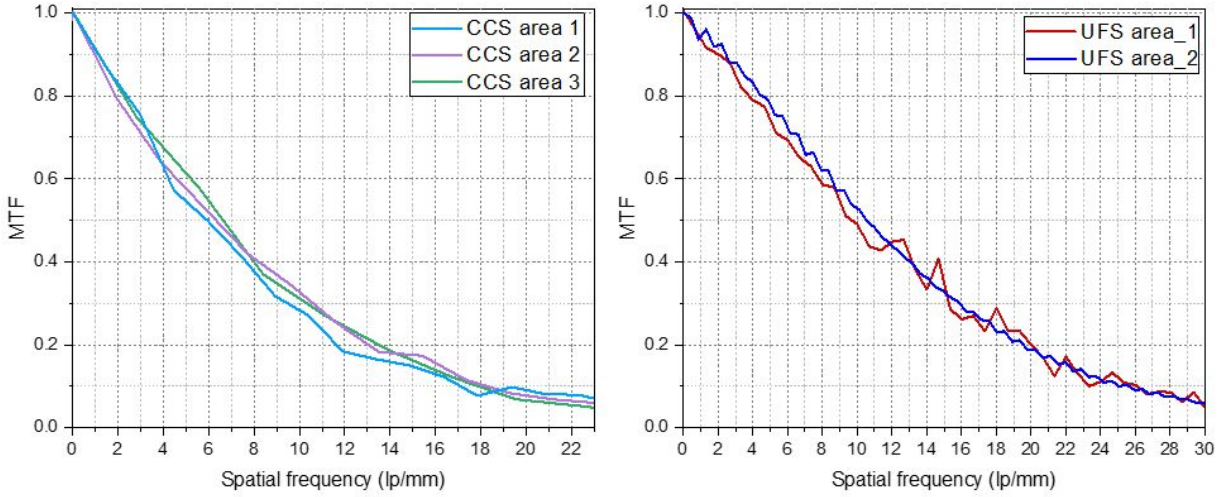


Figure S14. The MTF curves calculated for areas (same colors) on Figure S12 for on CCS (left) and UFS (right) screens.

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Light Yield as a Balance Between X-ray Absorption and Optical Losses

In practical measurements of scintillator light yield (LY), especially in layer-type geometries, the registered LY is not solely determined by the material's emission efficiency but rather results from the **interplay between X-ray absorption and internal optical losses**. This fundamental balance governs the shape of LY dependence on layer thickness and becomes especially critical when comparing materials with different geometries and scattering behavior.

When the total thickness of the scintillator layer is small, only a small portion of the incident X-ray energy is absorbed, and thus the number of generated scintillation photons is low. As the layer becomes thicker, **X-ray absorption increases**, leading to more photon generation and a rise in LY. However, beyond a certain thickness, **optical losses due to photon reabsorption and scattering** in the bulk begin to dominate. As photons must traverse longer paths through the material before escaping, the probability of absorption or deflection away from the detector increases, causing the registered LY to decline. This results in a **non-monotonic LY dependence on thickness**, with a clear maximum at a material- and geometry-specific optimal thickness.

To describe this behavior quantitatively, we introduce two complementary parameters:

- *h*: the **total thickness** of the scintillator layer, which includes both active and passive material (e.g., polymer matrix),
- *d*: the **scintillator-equivalent thickness (SET)**, representing the effective thickness of the active scintillating material within the layer.

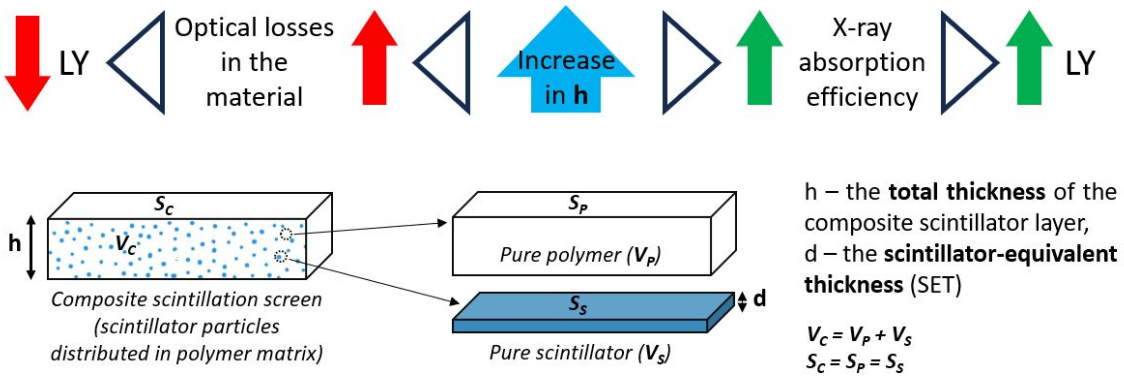


Figure S15. This schematic diagram illustrates the concept of the "apparent" light yield as a function of h scintillator-equivalent thickness (SET, d).

The parameter *d* is defined based on the mass loading and densities of the scintillator and matrix components. For composite scintillators composed of dispersed particles in a polymer, this formalism enables direct comparison of materials with different filler fractions and geometries, while decoupling X-ray absorption effects from those related to optical transport.

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The scintillator-equivalent thickness d is calculated as follows:

$$d = h \times [w_s/\rho_s]/\left[w_s/\rho_s + (1 - w_s)/\rho_p\right]$$

where h is the total scintillation screen thickness, w_s is the mass fraction of the scintillator in the composite, ρ_s and ρ_p being the densities of the scintillator and the polymer matrix, respectively.

This framework allows intuitive interpretation of light yield trends:

- Increasing d at fixed h enhances X-ray absorption,
- Increasing h at fixed d dilutes the concentration of scintillator, increases photon path length, and thus **worsens optical efficiency**.

This balance naturally explains the observed maxima in LY(d) dependencies and highlights the importance of describing composite systems in the (h, d) coordinate system rather than relying solely on mass fraction or total thickness.

Further details are available upon request from the corresponding author.

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Table 1. Comparison of main parameters of hybrid copper halide scintillators.

Formula	**Light yield, photons/MeV**	**Spatial resolution, l.p./mm**	**Decay time, us**	**Reference**
$(C_5H_5N)Cu_4I_4$	~44000	20	11.8	[2]
$[CuI(PPh_3)_2C_5H_4NCH_3]$	28400 ± 1300	9.8	24.8	[3]
$(C_3H_8N_2S)_3Cu_3Br_3$	20000 ± 700	11.7	1.3	[4]
$(C_{12}H_{11}N)Cu_4I_4$	36900	24.3		[5]
$(C_9H_{13}N)Cu_4I_4$	30800	21.2		[5]
$(C_8H_{20}N)_2Cu_2Br_4$	91300	9.5	56	[6]
$PPh_3C_2H_5CuBr_2$	58000	10	32.9	[7]
$(C_{10}H_{21}N_2)_2Cu_4I_6$	32600	31	5.58	[8]
$Cu_6I_6(HMTA)_2$	63500	18 (24.5)*	19.4**	This work

*in the form of flexible composite screen and ultrathin film screen on porous PVDF membrane (in parenthesis).

**according to TRPL

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Table S2. Comparison of apparent and XAE-corrected light yields of the composite scintillation screens of different d and h.

Sample	LY_a, photons/MeV	X-ray attenuation efficiency, %	$LY_{XAE-corr}$, photons/MeV
h = 75, d = 20	43220	57	76000
h = 75, d = 30	57940	68	85200
h = 75, d = 40	63530	75	84700
h = 75, d = 50	61150	87	70300

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